

# LipidIN × Agent — `code` User Manual (English)

Conversational, LLM-assisted pipeline for **untargeted lipidomics**: annotation → quantification → QC → differential analysis → biological conclusions → verified report.

A Chinese version of this manual is in [README\\_ZH.md](#) . For carbon–carbon double-bond localization (Paternò–Büchi), see the sibling folder [code\\_for\\_PB](#) .

This manual is written from an actual read of the whole pipeline and a real launch of the bundled demo. All screenshots below are from the live interface.

## 1. What this does

Given LC-MS/MS runs of your samples, the pipeline:

1. **Convert** vendor raw / mzML → standard pickles.
2. **Annotate** (MS2 library search against the LipidIN theoretical library).
3. **Retention-time** inference / alignment check.
4. Build the **master annotation table**.
5. **Quantify** (EIC extraction + peak integration).
6. **QC**: total-area correction, CV, PCA, PLS-DA.

The agent then adds differential analysis (volcano / heatmap / fatty-acid change / class bubble / random forest), pathway enrichment (LIPID MAPS), LLM-generated biological conclusions, a 4-axis verification pass (annotation / RT / biology / literature), and a bilingual Markdown report.

## 2. Requirements

| Item                                                         | Requirement                                                                                                                                                                                                           |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OS / Python                                                  | <b>Windows x64 + CPython 3.11</b> (the C++ engines ship as <code>*.cp311-win_amd64.pyd</code> )                                                                                                                       |
| RAM                                                          | ≥ <b>16 GB</b> (the theoretical libraries are large; use 1 search worker on 16 GB)                                                                                                                                    |
| Python packages                                              | <code>flask pandas numpy scipy scikit-learn matplotlib polars fisher_py</code>                                                                                                                                        |
| Thermo <code>.raw</code> reading                             | <code>fisher_py</code> + .NET (bundled)                                                                                                                                                                               |
| SCIEX <code>.wiff</code> /<br>Bruker/Agilent <code>.d</code> | ProteoWizard <b>msconvert</b> on PATH (auto-converted to mzML)                                                                                                                                                        |
| Spectral libraries                                           | <b>Download from Zenodo</b> → <a href="https://zenodo.org/records/21421866">https://zenodo.org/records/21421866</a> ( <code>pos_common.pkl</code> , <code>neg_common.pkl</code> ) into a <code>library/</code> folder |
| LLM (optional)                                               | An API key for Anthropic / OpenAI-compatible / DeepSeek. Structured commands work offline; only free-form requests, conclusions, and interpretation need it.                                                          |

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### 3. Prepare your data

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Organize inputs by polarity:

```
<data root>/
├ pos/      # positive-mode  A1.raw, A2.raw, ... , B1.raw, ... , (QC1, QC2, ... )
└ neg/      # negative-mode  A1.raw, ...
```

- Sample base names encode the group ( **A\*** vs **B\*** ); names containing **QC** are treated as pooled QC.
- Accepts **.mzML** , Thermo **.raw** , SCIEX **.wiff** , Bruker/Agilent **.d** .
- The bundled demo lives at **D:\bio\_inf\LipidIN\_github\data** ( **pos/** and **neg/** , 13 files each).

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### 4. Two ways to run

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#### Option A — One-shot batch ( **run.py** )

Open **run.py** and edit the three lines at the top:

```
POLARITY      = "pos"                                # "pos" or "neg"
DATA_ROOT     = r"D:\bio_inf\LipidIN_github\data"
LIBRARY_DIR   = r"D:\bio_inf\LipidIN_github\library"
```

Then:

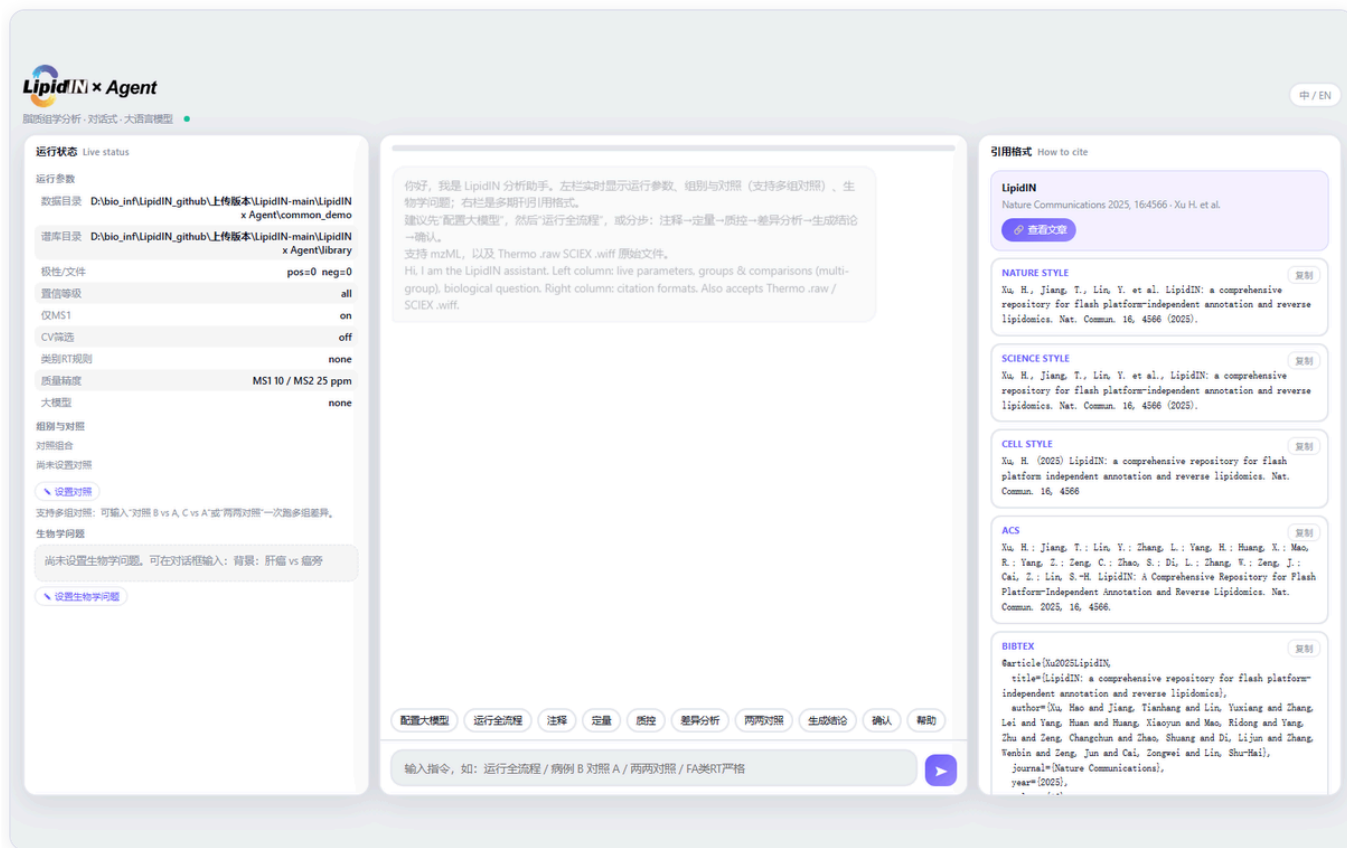
```
D:\anaconda3\python.exe run.py
```

It runs steps 1–6 for that polarity and writes **master\_table.csv** , **master\_table\_quant.csv** , and a **QC-<pol>/** folder. Keep **EQ3\_WORKERS = 1** on 16 GB. Run once per polarity.

#### Option B — Conversational agent (recommended)

```
D:\anaconda3\python.exe agent_server.py
# opens http://127.0.0.1:7860
```

Three-pane interface: **left** = live parameters, groups & comparisons, biological question; **center** = chat; **right** = citation formats.



## 5. A full standard conversation

Below is a real session against the demo.

### 5.1 Ask for help — 帮助 / help

The agent lists the commands: configure LLM, step runs ( `annotation` / `quantify` / `qc` / `differential` / `conclusions` / `confirm` ), `run all` , and how to change parameters (case/control, grade filter, CV cutoff, ppm/Da, biological background).

脂质组学分析 · 对话式 · 大语言模型

中 / EN

运行状态 Live status

运行参数

数据目录 D:\bio\_inf\LipidIN\_github\上传版本\LipidIN-main\LipidIN x Agent\common\_demo

数据库目录 D:\bio\_inf\LipidIN\_github\上传版本\LipidIN-main\LipidIN x Agent\library

极性/文件 pos=0 neg=0

置信等级 all

仅MS1 on

CV筛选 off

类别RT规则 none

质量精度 MS1 10 / MS2 25 ppm

大模型 none

脂组与对照

对照组合

尚未设置对照

设置对照

支持多组对照: 可输入“对照 B vs A, C vs A”或“两两对照”一次跑多组差异。

生物学问题

尚未设置生物学问题。可在对话框输入: 背景: 肝癌 vs 瘤旁

设置生物学问题

你好, 我是 LipidIN 分析助手。左栏实时显示运行参数、组别与对照 (支持多组对照)、生物学问题; 右栏是多期刊引用格式。

建议先“配置大模型”, 然后“运行全流程”, 或分步: 注释→定量→质控→差异分析→生成结论→确认。

支持 mzML, 以及 Thermo.raw SCIEX.wiff 原始文件。

Hi, I am the LipidIN assistant. Left column: live parameters, groups & comparisons (multi-group), biological question. Right column: citation formats. Also accepts Thermo.raw / SCIEX.wiff.

帮助

我可以运行整个脂质分析流程。建议第一步先『配置大模型』, 配置后即可用自然语言下达要求。

配置大模型: 回复“配置大模型”, 我会带你填写 提供方/密钥/模型

分步运行: 注释 (仅定性) / 定量 (仅定量) / 质控 (仅质控) / 差异分析 (仅差异分析) / 生成结论 / 确认 (验证并出报告)

运行全流程: 定性→定量→质控→差异分析→生物学结论 一次跑完

改参数: 病例 B 对照 A、等级 1,2、排除 ms1-only、cv 25、ms2 ppm 20、背景: 肝癌 vs 瘤旁

个性化 (配置大模型后): 如“FA类RT严格, 其它类别宽松”

每次运行前我都会让你确认目录/极性/脂库/分组等参数; 随时可改参数并重复某一步

配置大模型

运行全流程

注释

定量

质控

差异分析

两两对照

生成结论

确认

帮助

输入指令, 如: 运行全流程 / 病例 B 对照 A / 两两对照 / FA类RT严格

引用格式 How to cite

LipidIN

Nature Communications 2025, 16:4566 · Xu H. et al.

查看文章

NATURE STYLE

Xu H., Jiang T., Lin Y. et al. LipidIN: a comprehensive repository for flash platform-independent annotation and reverse lipidomics. Nat. Commun. 16, 4566 (2025).

复制

SCIENCE STYLE

Xu H., Jiang T., Lin Y. et al., LipidIN: a comprehensive repository for flash platform-independent annotation and reverse lipidomics. Nat. Commun. 16, 4566 (2025).

复制

CELL STYLE

Xu H. (2023) LipidIN: a comprehensive repository for flash platform-independent annotation and reverse lipidomics. Nat. Commun. 16, 4566

复制

ACS

Xu H.; Jiang, T.; Lin, Y.; Zhang, L.; Yang, H.; Huang, X.; Mao, R.; Yang, Z.; Zeng, C.; Zhao, S.; Di, L.; Zhang, W.; Zeng, J.; Cai, Z.; Lin, S.-H. LipidIN: A Comprehensive Repository for Flash Platform-Independent Annotation and Reverse Lipidomics. Nat. Commun. 2025, 16, 4566.

复制

BIBTEX

@article{Xu2025LipidIN, title={LipidIN: a comprehensive repository for flash platform-independent annotation and reverse lipidomics}, author={Xu, Hao and Jiang, Tianshang and Lin, Yunxiang and Zhang, Lei and Yang, Huan and Huang, Xiaoyun and Mao, Ridong and Yang, Zhu and Zeng, Changchun and Zhao, Shuang and Di, Lijun and Zhang, Wenbin and Zeng, Jun and Cai, Congwei and Lin, Shu-Hai},

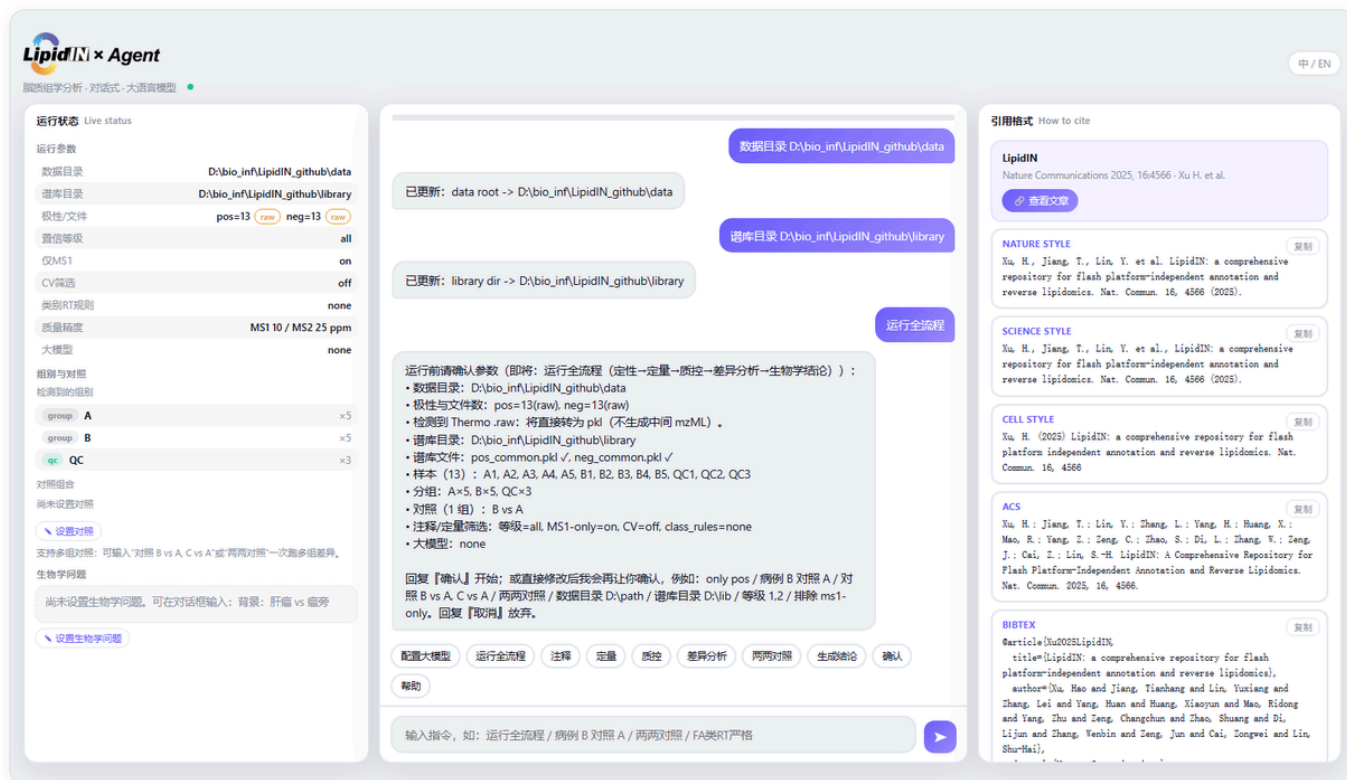
复制

## 5.2 Point it at your data & library

data folder D:\bio\_inf\LipidIN\_github\data  
library folder D:\bio\_inf\LipidIN\_github\library

## 5.3 Run everything — 运行全流程 / run all

The agent **shows a preflight summary and waits for your confirmation** — polarity/file counts, detected samples & groups, library check, and current filters — before doing any heavy work. Reply **确认** / **confirm** to start, or edit any parameter first.



As shown, it detected `pos=13`, `neg=13`, samples `A1..B5`, `QC1..QC3`, groups `A×5`, `B×5`, `QC×3`, comparison `B vs A`, and verified `pos_common.pkl ✓` `neg_common.pkl ✓`.

## 5.4 Change parameters mid-conversation

Any parameter can change at any time and only the affected downstream stage re-runs:

- Comparison: `case B control A`, or `B vs A`, `C vs A`, or `all pairs`.
- Confidence grade filter: `grade 1,2` (Level 1 = MS2 with all diagnostic fragments; 2 = MS2 with a missing fragment; 3 = MS1-inferred).
- `exclude ms1-only`, `cv 25`, `ms2 ppm 20`.
- Per-class RT strictness (needs LLM): *"keep RT strict for FA, loose for others"*.
- Biological question: `background: liver tumor vs adjacent`.

## 5.5 Conclusions & report

`generate conclusions` → the agent drafts biological conclusions; `confirm` runs the 4-axis verification and writes `LipidIN_report.md` (annotation results, QC, differential analysis + pathways, conclusions with a verification table, appendix).

## 6. What you get (outputs)

| File / folder                                                     | Meaning                                                        |
|-------------------------------------------------------------------|----------------------------------------------------------------|
| <code>&lt;pol&gt;/pkl/master_table.csv</code>                     | Master annotation table (species, evidence type, RT judgement) |
| <code>&lt;pol&gt;/pkl/master_table_quant.csv</code>               | Per-sample quantities (EIC areas)                              |
| <code>QC_&lt;pol&gt;/</code>                                      | Total-area-corrected table + CV / PCA / PLS-DA plots           |
| <code>differential_*.csv</code> + PNGs                            | Volcano / heatmap / fatty-acid / class-bubble / random-forest  |
| <code>pathway_enrichment_*.png/.csv</code>                        | LIPID MAPS pathway enrichment                                  |
| <code>&lt;data<br/>root&gt;/agent_report/LipidIN_report.md</code> | Final bilingual report                                         |

## 7. Troubleshooting (anticipated errors)

### ① `ImportError: eq3_search / eq3_similarity (C++ engine won't load)`.

The engines are prebuilt for **CPython 3.11, Windows x64** ( `*.cp311-win_amd64.pyd` ). A different Python version/platform fails the import. Fix: run with **Python 3.11 on Windows x64**. `eq3_similarity` can be rebuilt from its source ( `build_cpp.py / compile_cpp_similarity.py` , needs MSVC + pybind11); `eq3_search` has **no source in this folder** ( `compile_cpp_fixed.py` even points at a hardcoded `EQ3.cpp` path that isn't present), so keep the shipped `.pyd` and match Python 3.11.

② **Library file not found.** `pos_common.pkl / neg_common.pkl` are **not** in the repo (too large). Download from <https://zenodo.org/records/21421866> into `library/` and point `LIBRARY_DIR` / the agent at it.

③ **MemoryError / heavy swapping.** The libraries need  $\geq 16$  GB. Keep `EQ3_WORKERS = 1` on a 16 GB machine; run one polarity at a time.

④ **`.wiff / .d` not processed / "msconvert not found"**. Non-Thermo vendor formats are converted via ProteoWizard **msconvert** — install it and ensure it's on PATH. Thermo `.raw` is read directly (no msconvert needed) but requires the `fisher_py` .NET backend.

⑤ **A path with a space is truncated in chat.** The agent stops a typed path at the first space. Use a **space-free** data/library path, or set paths by editing `run.py` .

⑥ **Wrong or empty groups.** Groups come from sample base names ( `A*` vs `B*` , `QC*` = pooled QC). If detection is wrong, set it explicitly: `case B control A` . Empty means no files were found under `pos/ neg/` — check the folder layout.

⑦ **Port already in use.** Another process holds `7860` . Edit the port near the bottom of `agent_server.py` , or free the port.

⑧ **Garbled console / UnicodeEncodeError** . Entry points force UTF-8; if you invoke a module directly on a GBK console you may see mojibake — run via `run.py / agent_server.py` or set `PYTHONIOENCODING=utf-8` .

⑨ **LLM features do nothing.** Without configuring a provider, structured commands still run; free-form requests, per-class RT rules, biological conclusions, and interpretation need an LLM key.

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## 8. Citation

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LipidIN — *Nature Communications* 2025, 16:4566 (Xu, H. et al.). Spectral libraries:  
<https://zenodo.org/records/21421866>